

HOMEWORK ASSIGNMENT 3 – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

Instructions:

- Kindly recall that homework assignments for this course are **not turned in**, and **not graded**.
 - However, problems from homework sets will form the basis of problems that appear on the weekly quizzes; as such, it is in your best interest to complete the homework sets on your own
 - Solutions to this assignment will be released at 8:00am on Friday, September 11 in time for the corresponding quiz on **Monday September 14**.
 - **Kindly Note:** Quiz 03 will be the full 15 questions and last for the full 20 minutes. 10 questions will be based off this week's material; 5 questions will be based off last week's material (to compensate for the fact that Quiz 02 did not take place as scheduled).
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1. **Is This Die Balanced?** I have recently acquired a six-sided die that, when rolled 12 times, landed on an even number (i.e. 2, 4, or 6) ten times. With this information, I'd like to investigate whether or not the die is balanced (has an equal chance of landing on an even or odd number) or loaded in favor of even numbers (has a higher chance of landing on an odd number than an even one).

- (a) Describe an observational unit in the context of this dataset.

Solution: One observational unit is **one trial of rolling the die and observing whether the outcome is even or not**.

- (b) Define the parameter of interest.

Solution: The parameter of interest is the **long-run proportion of times the die lands on an even number**.

- (c) Calculate the value of the statistic.

Solution: The value of the statistic is the sample proportion of times the die landed on an even number: **$10/12 = 0.8333$** .

- (d) Consider the variable: "whether a particular roll landed on an even number or not." What are the possible levels of this variable, and which level are you considering a "success?"

Solution: Since the die will either land on an even number or not, the levels of the variable are either "**yes**" or "**no**;" an equivalent encoding is "**even**" and "**odd**." We are considering a "yes" (or "even") value to be a "success."

A Note: At this stage in the course, there is some flexibility in what we consider a "success," and, in general, you won't run into any problems as long as you're consistent. For now, though, we'll try and match "success" to the research conjecture as much as possible - since our conjecture has something to do with whether or not the die is balanced in favor of **even** numbers, we'll pick "even" to be a success.

- (e) State the research question in two ways: one verbally (using words and only minimal math), and one mathematically (involving the parameter of interest you defined above).

Solution: The research question is: **is the die balanced, or loaded in favor of even numbers?** In terms of the parameter of interest, this boils down to: is the value of the parameter **equal to 0.5** (indicating a balanced die) or **greater than 0.5** (indicating a bias in favor of even numbers)?

- (f) Fill in the entries in the right-most column of the table below, to set up the chance model for this study.

One Coin Flip	One roll of the die
Coin Landing Heads	The die landing on an even number
Tails	The die landing on an odd number
Chance of Heads	0.5 (indicating a balanced die)
One Repetition	One set of 12 simulated rolls of the die

Solution: Note that we want “Coin landing Heads” to match our notion of a “success.”

- (g) Use the One Proportion Applet to simulate 2,000 repetitions of the chance model. Include a screenshot or sketch of your plot below. **Don’t close out of the applet yet - the next question will ask you to modify your plot!**

Solution: Answers may vary; here is a sample plot:

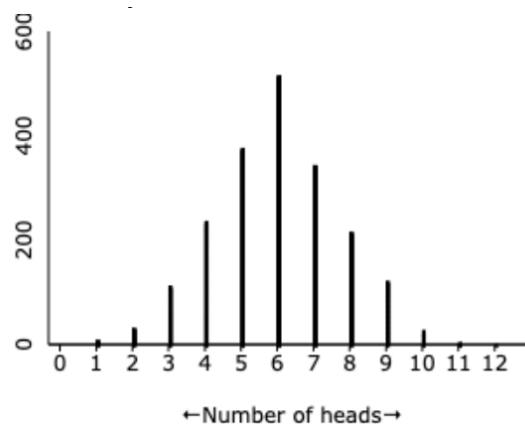


Figure 1: A sample plot from the applet.

- (h) Provide an approximate value for the probability that ten out of twelve rolls of a balanced six-sided die land on an even number. Do this by:

1) Navigating back to the applet, and find the line that says “As extreme as.”

- 2) Clicking the $\geq$ symbol twice until it turns into the $=$ symbol, and entering the appropriate value (it's up to you to identify the correct value to put based on the problem statement!) in the provided blank.

Again, do not close out of the applet as we will modify it further in the next part of this question.

Solution: Remember that probabilities are long-run proportions, meaning we can approximate them by examining large-sample proportions. In particular, a good approximation to the probability that ten-of-twelve rolls of a balanced six-sided die landed on an even number is the proportion of times, among the 2,000 repetitions of twelve rolls of a balanced six-sided die, we observed exactly ten even numbers. From the plot, this should be around $28 / 2000 = 0.0140$, though your exact value will depend on what your particular plot from the previous part looks like.

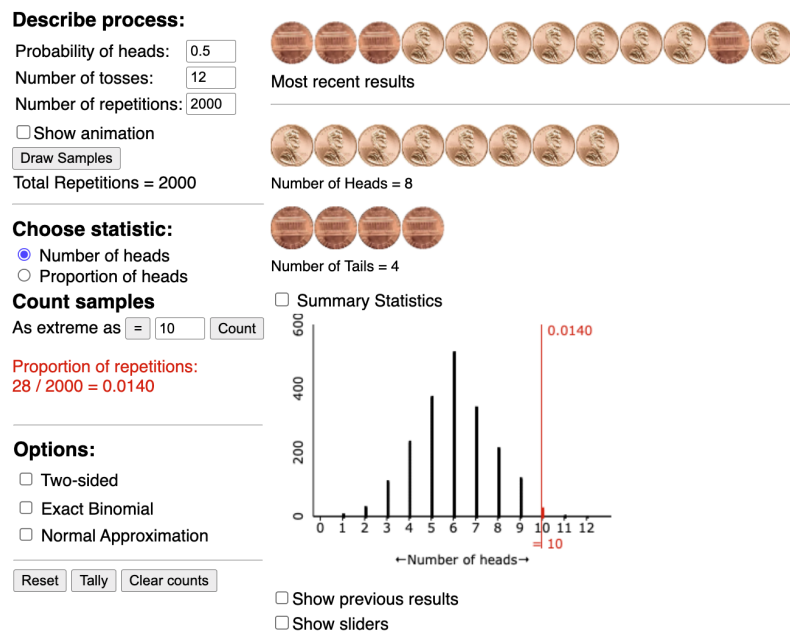


Figure 2: A sample plot from the applet.

If you're curious, the exact probability value (computed using mathematical tools that are outside the scope of this course) is 0.0161.

- (i) Explain why the value you obtained in part (g) is just an *approximation* to the desired probability.

Solution: Recall that a probability is a long-run proportion; i.e. it essentially the proportion of times we would expect to see an event occur in an infinite number of repetitions. Though 2,000 is a large number of repetitions, it is still not infinite.

- (j) Provide an approximate value for the probability that ten or more rolls out of a total of twelve rolls of a balanced six-sided die land on an even number. Do this by:

- 1) Navigating back to the applet, and going back to the line that says "As extreme as."
- 2) Clicking the $=$ symbol until it turns into the $\geq$ symbol (make sure you still have the value you entered before in the provided blank space).

Solution: From my plot, I got a value of around $34 / 2000 = 0.0170$, though, again, your exact value will depend on what your particular plot from the applet looks like.

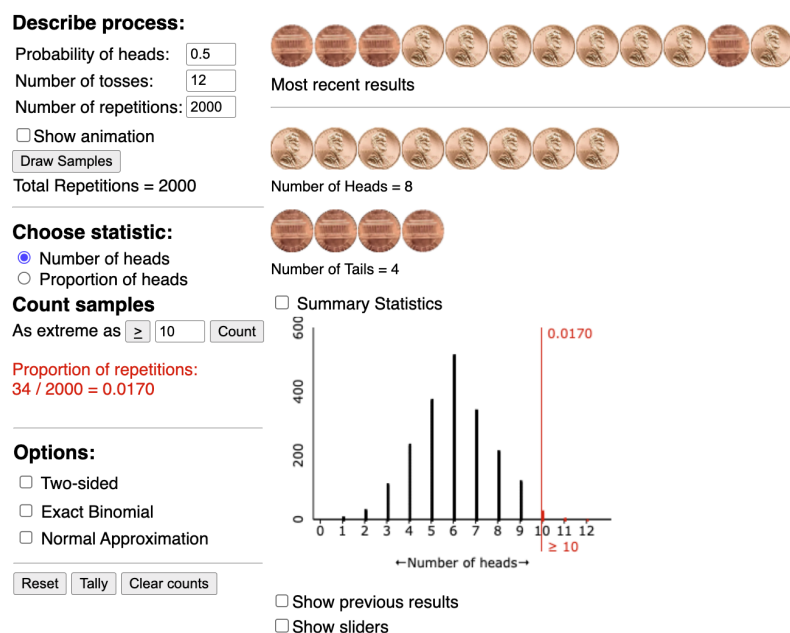


Figure 3: A sample plot from the applet.

- (k) The true/exact probability that ten or more out of twelve rolls of a balanced six-sided die land on an even number can be computed (using mathematical tools that are outside the scope of this course) to be 0.0192. Provide an interpretation of this value.

Solution: In the long-run, we'd expect the proportion of times twelve rolls of a balanced six-sided die lands on an even number ten or more times to be around 0.0192.

- (l) Is it likely that ten or more of twelve rolls of a balanced six-sided die land on an even number? What does this mean about the evidence for/against the possibility that the die is balanced, vs. the possibility that the die is biased in favor of even numbers?

Solution: It's fairly uncommon to observe ten or more even numbers among twelve rolls of a perfectly balanced die. As such, there seems to be evidence that the die is **not balanced** and instead **biased in favor of even numbers**.

Note: you may ask why we're including "ten or more" instead of just looking at ten - we'll revisit this during our lecture on Monday.

2. **Cat Loafs.** The term "loafing" is commonly used among cat owners to describe the act of a cat sitting in a particular way. When "loafed," a cat can either curl their tail clockwise or counterclockwise. Suppose our research question is: are cats more likely to curl their tails clockwise when loafing? I (Dr. Ethan) conducted a (very rudimentary) study in which I watched a few videos on *YouTube* of cats loafing; among the 23 loafing cats I observed, 14 of them curled their tails clockwise (and the remainder curled their tails counterclockwise). The particular research conjecture we'll consider is: do cats prefer to curl their tails clockwise when loafed?

- (a) Define the parameter of interest.

Solution: The parameter of interest is the **long-run proportion of times a cat will choose to curl their tail clockwise when loafed.**

- (b) Calculate the value of the statistic

Solution: The value of the statistic is **$14/23 \approx 0.6087$**

- (c) Fill in the entries in the right-most column of the table below, to set up the chance model for this study.

One Coin Flip	One cat loafing and choosing which direction to curl its tail
Coin Landing Heads	A cat choosing to curl its tail clockwise when loafed
Tails	A cat choosing to curl its tail clockwise when loafed
Chance of Heads	0.5 (indicating no preference for clockwise over counter-clockwise)
One Repetition	One set of 23 simulated cat loafs

- (d) Shown below is a plot from the One Proportion Applet representing 1,000 repetitions of the chance model. Also indicated in the plot is the fact that around 20% of the 1,000 repetitions corresponded to samples with sample proportions of 14/23 or greater.

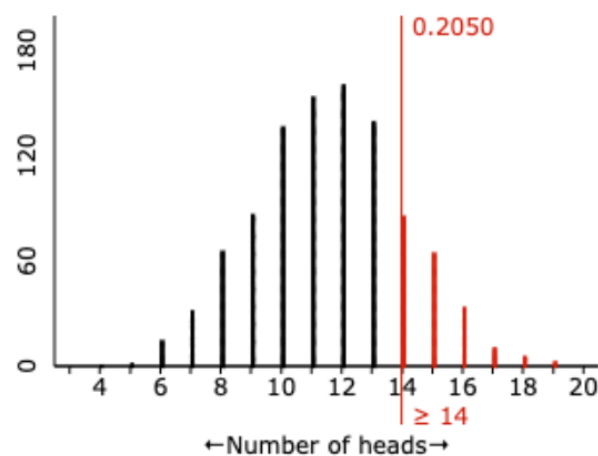


Figure 4: 1,000 repetitions, as conducted by the One Proportion applet.

This shows that it is not that unlikely to observe a sample proportion of 14/23 or greater. Does this **prove** that cats prefer to curl their tails clockwise when loafed? If not, how would you phrase the conclusions of the study, based on the simulation of the chance model?

Solution: No, this does not “prove” that cats prefer to curl their tails clockwise when loafed; all it says is that our data is **not inconsistent** with the research conjecture, that cats prefer the clockwise direction over the counterclockwise direction.

What we can say is that “given the data we observed, the research conjecture is plausible.” Remember - we never get to know/prove whether the conjecture is true or not! This is a point we’ll return to periodically, including on Monday when we continue our discussion on performing inference on a single proportion.